LESSON 6: ANTIHYPERTENSIVE DRUGS

Blood pressure, hypertension, and the 4 drug groups by where they act

How do I keep the antihypertensives straight?

Start from **BP = CO × SVR**. Every drug lowers **cardiac output**, **vascular resistance** or **blood volume** at one of **4 sites**: **arterioles**, **venules**, **heart**, **kidneys**. The slides group them into **4 classes**: **diuretics** (kidney, volume), **sympatholytics** (block the sympathetic push), **direct vasodilators** (open the vessels) and **angiotensin blockers** (stop the RAAS). The **name endings** give away most classes.

CIRCULATION REVIEW

- **Blood flow**: body → **right heart** → **lungs** (pulmonary circuit) → **left heart** → body (systemic circuit)
- **Arteries** carry blood **away from** the heart; **veins** carry it **back** and have **valves**; **capillaries** = one layer of **endothelial cells** (gas exchange)
- Vessel wall layers: **tunica interna** (endothelium, subendothelial layer, internal elastic lamina) → **tunica media** (smooth muscle, external elastic lamina) → **tunica externa**

BLOOD PRESSURE

- **Blood pressure** – the pressure exerted by blood on the **walls of a blood vessel**
- **BP = CO × SVR** (cardiac output × systemic vascular resistance)
- **CO = HR × SV** (heart rate × stroke volume)
- **Systolic (SBP)** – the **upper** number (150 in 150/90); pressure while the heart is **contracting** (at the heartbeat)
- **Diastolic (DBP)** – the **lower** number (90); pressure while the heart is **relaxed** (between beats)

WHAT RAISES MEAN ARTERIAL PRESSURE (MAP)

- ↑ **Cardiac output**, from:
  - ↑ **Heart rate** – less parasympathetic, more **sympathetic** impulses and **adrenal medulla** hormones
  - ↑ **Stroke volume** – more **venous return** (more blood volume, skeletal muscle pump, respiratory pump, venoconstriction)
- ↑ **Systemic vascular resistance**, from:
  - ↑ **Blood viscosity** (more RBCs, as in **polycythemia**)
  - ↑ **Total vessel length** (bigger body size, as in **obesity**)
  - ↓ **Vessel radius** (**vasoconstriction**)

HORMONES THAT CONTROL BP

Note: Dr. Orig **skipped** this table and the 4-sites figure below in class and said to **focus on the drugs**. They are kept here for reference.

- **Raise heart rate and contractility** (↑ BP): **norepinephrine, epinephrine**
- **Vasoconstrict** (↑ BP): **angiotensin II, ADH**, norepinephrine (α1), epinephrine
- **Vasodilate** (↓ BP): **ANP**, epinephrine (β2, in heart and skeletal muscle arterioles), **nitric oxide**
- **Raise blood volume** (↑ BP): **aldosterone, ADH**
- **Lower blood volume** (↓ BP): **ANP** (atrial natriuretic peptide)

Note: Memory tip: ANP is the only hormone on the list that **lowers** BP in two ways (vasodilation and less volume). **ADH** and **angiotensin II** raise it in two ways.

4 ANATOMIC SITES OF BP CONTROL

- **1. Arterioles** – **resistance** vessels
- **2. Venules** – **capacitance** vessels
- **3. Heart** – **pump output**
- **4. Kidneys** – **volume** (through **renin** → **angiotensin** → **aldosterone**)
- All four are driven by the **CNS sympathetic nerves**

HYPERTENSION

- Diagnosed on **repeated, reproducible** measurements of elevated BP, not one reading

- **Primary (essential)** – **no specific cause** (most patients)
- **Secondary** – has a **specific etiology**

BP CATEGORIES (SYSTOLIC / DIASTOLIC, MMHG)

- **Normal** – below **120 and** below **80**
- **Elevated** – **120–129 and** below 80
- **Stage 1** – **130–139 or 80–89**
- **Stage 2** – **≥140 or ≥90**
- **Hypertensive crisis** – above **180 and/or** above **120**

Note: Only "Normal" and "Elevated" need **both** numbers in range (**and**); stages 1 and 2 need **either** number (**or**). So 125/85 is **Stage 1** because of the diastolic.

THE 4 CLASSES OF ANTIHYPERTENSIVES

- **1. Diuretics**
- **2. Sympathoplegic (sympatholytic) agents**
- **3. Direct vasodilators**
- **4. Agents that block the production or action of angiotensin**

1. DIURETICS

- Lower BP by **depleting body sodium stores**
- Lower BP by **10–15 mmHg** in most patients
- Used **alone** in **mild or moderate essential hypertension**

SITES OF ACTION (ALONG THE NEPHRON)

- **1. Proximal convoluted tubule** – **acetazolamide** (carbonic anhydrase inhibitor)
- **2. PCT and descending limb** – **osmotic diuretics** (**mannitol**)
- **3. Thick ascending limb (TAL) of the loop of Henle** – **loop diuretics** (furosemide, torsemide)
- **4. Distal convoluted tubule** – **thiazide-type** (HCTZ, chlorthalidone, metolazone)
- **5. Collecting duct** – **potassium-sparing** (spironolactone, triamterene)

Note: Memory tip: go down the nephron in **A-O-L-T-K** order: "**A**ll **O**ur **L**oops **T**ake **K**+": **A**cetazolamide, **O**smotic, **L**oop, **T**hiazide, **K**-sparing.

LOOP DIURETICS (THE STRONGEST)

- Inhibit **NaCl reabsorption in the TAL**
- **MOA:** inhibit **NKCC2**, the luminal **Na+/K+/2Cl- cotransporter** in the TAL
- **2 prototypes: furosemide and ethacrynic acid**; also **bumetanide, torsemide**
- **Uses: acute pulmonary edema** and other edematous conditions; **hyperkalemia**; **acute renal failure**
- **Toxicity: ototoxicity**; **allergic reactions** (sulfa)
- Daily oral doses: **bumetanide 0.5–2 mg**, **torsemide 5–20 mg**, **furosemide 20–80 mg**, **ethacrynic acid 50–200 mg**

Note: Memory tip: the **ear has a loop too** (the cochlea), so **loop** diuretics hurt the **ear** (ototoxicity). Loops **throw out K+**, which is why they treat **hyperkalemia**.

Note: Textbook extra: **ethacrynic acid** is the loop diuretic **without a sulfa group**, the one for patients allergic to sulfa drugs.

THIAZIDE DIURETICS

- **MOA:** block the **Na+/Cl- cotransporter (NCC)** in the **DCT**
- **Prototype: hydrochlorothiazide (HCTZ)**, 25–100 mg once a day
- **Uses: hypertension**, **heart failure**
- **Toxicity: hyperlipidemia**, **hyponatremia**, **gout**, **allergic reactions**
- **Thiazide-like** (not thiazides, but **sulfonamides** that act the same): **chlorthalidone, indapamide, metolazone, quinethazone**
- Taken in **two divided doses**: **chlorothiazide** (0.5–2 g) and **hydroflumethiazide**; all others **once a day**

Note: Memory tip: thiazide toxicity = **LUNA**: **L**ipids up, **U**ric acid up (gout), **Na+** down, **A**llergy (sulfa).

Note: Loop vs thiazide: loop = **NKCC2** in the **TAL** (strongest, for **edema**); thiazide = **NCC** in the **DCT** (for **hypertension**).

POTASSIUM-SPARING DIURETICS

- Prevent **K+ secretion** by antagonizing **aldosterone** in the **collecting tubules**
- **1. Aldosterone antagonists** – **spironolactone, eplerenone** (block the mineralocorticoid receptor)

- **2. Epithelial Na+ channel (ENaC) inhibitors** – **amiloride, triamterene**
- **Uses: hyperaldosteronism**, **heart failure**
- **Toxicity: hyperkalemia**; **gynecomastia (spironolactone)**

Note: Memory tip: they **spare K+**, so K+ builds up: **hyperkalemia**. **Spiro**nolactone is a steroid look-alike, so it causes **gynecomastia**; **eplerenone** is the more selective one.

2. SYMPATHOLYTIC (SYMPATHOPLEGIC) DRUGS

- Sympathetic discharge → **α1** on **blood vessels** (vasoconstriction) + **β1** on the **heart** (↑ rate and force, ↑ afterload) → ↑ BP. These drugs cut that push.
- **1. Methyldopa**
- **2. Clonidine**
- **3. Beta-blockers**
- **4. Prazosin** and other **alpha-1 blockers**
- **5. Phentolamine, phenoxybenzamine**

Note: From Lesson 4: **methyldopa** and **clonidine** are **central α2 agonists** (the brain lowers sympathetic outflow). **Methyldopa** = **M**om's drug (**gestational hypertension, preeclampsia**). **-olol** = β-blockers; **-zosin** = α1 blockers; **phe**ntolamine and **phe**noxybenzamine are for **phe**ochromocytoma.

Note: Textbook extra: stopping **clonidine** suddenly causes **rebound hypertension**.

3. DIRECT VASODILATORS

- **Oral: hydralazine**, **minoxidil**
- **Parenteral (IV): nitroprusside**, **fenoldopam**

MECHANISMS (KATZUNG TABLE 11-3)

- **Release nitric oxide** (from the drug or the endothelium) – **nitroprusside, hydralazine**, nitrates, histamine, acetylcholine
- **Reduce calcium influx** – **verapamil, diltiazem, nifedipine** (the CCBs)
- **Hyperpolarize the membrane by opening K+ channels** – **minoxidil, diazoxide**
- **Activate dopamine receptors** – **fenoldopam**

Note: Memory tip: Minoxidil and **Dia**zoxide **open the K+ door**. **Fenol-DOPA-m** works on **DOPA**mine receptors. The slide shows **topical minoxidil 5%** (Regroe): its side effect of **hair growth** became a **hair-regrowth** product.

CALCIUM CHANNEL BLOCKERS (CCBS)

- **MOA:** inhibit **Ca2+ influx into arterial smooth muscle** cells → **dilated arteries**, lower BP
- **Non-dihydropyridines: verapamil, diltiazem** – also **slow the SA node** (heart rate), so the ventricle fills fully and the heart's workload drops
- **Dihydropyridines (-dipine): amlodipine, felodipine, nifedipine, nicardipine** – mainly the **vessels**

Note: Memory tip: "**-dipine dilates**" (vessels). **V**erapamil and **D**iltiazem also **V**isit the heart and **D**ecrease its rate.

4. INHIBITORS OF ANGIOTENSIN (RAAS)

THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

- Trigger: **drop in BP** or **drop in fluid volume**
- **Kidney** releases **renin** → renin turns **angiotensinogen** (from the **liver**) into **angiotensin I**
- **ACE** (angiotensin-converting enzyme, from the **lungs**) turns angiotensin I into **angiotensin II**
- **Angiotensin II** → **vasoconstriction** (narrows vessels directly) + **adrenal gland releases aldosterone**
- **Aldosterone** → kidneys **reabsorb salt (NaCl) and water** → more volume, higher BP

WHERE EACH DRUG HITS THE RAAS

- **Aliskiren** – blocks **renin**
- **ACE inhibitors** – block **ACE** (Ang I → Ang II)
- **ARBs** – block the **angiotensin II receptor** (both vasoconstriction and aldosterone release)
- **Spironolactone, eplerenone** – block **aldosterone's** effect
- **Bradykinin** side branch: **ACE is the same enzyme as kininase II**, which breaks down **bradykinin** (a **vasodilator** that also raises **prostaglandins**)

Note: Memory tip: go down the cascade **R-A-A-A**: **R**enin (ali**skiren**) → **A**CE (-**pril**) → **A**ngiotensin II receptor (-**sartan**) → **A**ldosterone (spironolactone).

ACE INHIBITORS (-PRIL)

- **MOA:** inhibit the enzyme that **converts angiotensin I to angiotensin II** and **inactivates bradykinin**, so **Ang II falls** and **bradykinin rises**
- **Drugs: captopril, enalapril, ramipril**
- Used in **chronic kidney disease**: they **diminish proteinuria** and **stabilize renal function** (especially in **diabetics**)
- **Toxicity: dry cough**, **angioedema** (swollen lips, shown on the slide), **contraindicated in pregnancy**

Note: Memory tip: the extra **bradykinin** causes the **cough** and the **angioedema**. ACEi side effects = **ACE**: **A**ngioedema, **C**ough, **E**mbryo (no pregnancy).

ANGIOTENSIN RECEPTOR BLOCKERS (ARBS, -SARTAN)

- **MOA:** block the angiotensin II **type 1 (AT1)** receptor
- **Drugs: losartan, valsartan, candesartan, irbesartan, telmisartan**
- **More selective** than ACE inhibitors: **no effect on bradykinin**
- **AT1** effects (blocked): **vasoconstriction, sympathetic activation, cell proliferation, aldosterone release, renal sodium reabsorption** → atherosclerosis, hypertension
- **AT2** effects (left alone): **vasodilation, inhibition of cell growth, apoptosis**

Note: ACEi vs ARB: same result on angiotensin, but the ARB leaves **bradykinin** alone, so **no cough**. A patient with an ACEi cough is switched to an **ARB**. Both are **contraindicated in pregnancy**.

TREATMENT ALGORITHM (ADULTS)

- **BP ≥ 140/90** → advise **therapeutic lifestyle changes**: **low-salt diet, weight loss, exercise, smoking cessation, alcohol abstinence**
- **High risk?** (**HMOD**, CV disease, or CV-risk equivalent)
  - **Yes** → start drugs now
  - **No** → lifestyle for **3 months**; target reached → keep the lifestyle change; not reached → start drugs
- **First-line drugs** (alone or combined): **ACE inhibitor, ARB, CCB, thiazide diuretic**
- Targets not met after **3 months**:
  - On **monotherapy** → add a **2nd** first-line agent
  - Already on **2 drugs** → add **another** first-line drug
  - Still not at target on the combination → **refer**
- **Preferred combinations: ACEi/ARB + CCB** and **ACEi/ARB + HCTZ**
- **Beta-blockers** as **initial** therapy only with **coronary artery disease, acute coronary syndrome, high sympathetic drive**, or **pregnant women**
- **Resistant hypertension** – still uncontrolled on **3 drugs, one of them a diuretic** → refer to a **specialist** for work-up or **second-line** agents

Note: Memory tip: first line is **A-C-D**: **A**CEi or **A**RB, **C**CB, **D**iuretic (thiazide). **B** (beta-blocker) waits for a reason: a sick heart, too much sympathetic drive, or pregnancy. Never combine an **ACEi with an ARB** (both are "A").

Name endings at a glance

-thiazide thiazide · **-semide / -etanide** loop · **-pril** ACE inhibitor · **-sartan** ARB · **-kiren** renin inhibitor · **-dipine** CCB (dihydropyridine) · **-olol** β-blocker · **-zosin** α1 blocker

SUMMARY TABLE: ANTIHYPERTENSIVE DRUGS

Class (ending)	Drugs	Mechanism (site)	Uses	Toxicity
Loop diuretics (-semide, -etanide)	**furosemide**, **ethacrynic acid** (prototypes), **bumetanide**, **torsemide**	block **NKCC2** (Na+/K+/2Cl-) in the **TAL**	**acute pulmonary edema**, edema, **hyperkalemia**, **acute renal failure**	**ototoxicity**, **allergy** (sulfa)
Thiazides (-thiazide)	**HCTZ** (prototype); thiazide-like: **chlorthalidone, indapamide, metolazone**	block **NCC** (Na+/Cl-) in the **DCT**	**hypertension**, **heart failure**	**hyperlipidemia, hyponatremia, gout, allergy** (LUNA)
K+-sparing: aldosterone antagonists	**spironolactone**, **eplerenone**	block aldosterone in the **collecting tubule**	**hyperaldosteronism**, **heart failure**	**hyperkalemia**; **gynecomastia** (spironolactone)
K+-sparing: ENaC inhibitors	**amiloride**, **triamterene**	block the epithelial Na+ channel (**ENaC**)	with other diuretics to spare K+	**hyperkalemia**
Central α2 agonists	**methyldopa**, **clonidine**	brain lowers sympathetic outflow	hypertension; **methyldopa in pregnancy**	clonidine: **rebound hypertension** if stopped
β-blockers (-olol)	propranolol, **atenolol**, **metoprolol**, labetalol…	block **β1** → ↓ heart rate and force	first line only with **CAD, ACS, high sympathetic drive, pregnancy**	bradycardia; bronchospasm (nonselective)
α1 blockers (-zosin)	**prazosin**, doxazosin	block α1 → vessels relax	hypertension, BPH	**orthostatic hypotension**
α nonselective	**phentolamine**, **phenoxybenzamine**	block α1 + α2	**pheochromocytoma**	reflex tachycardia
Oral vasodilators	**hydralazine** (NO), **minoxidil** (K+ channel opener)	relax arteriolar smooth muscle	hypertension	minoxidil: **hair growth** (now a topical hair-regrowth drug)
Parenteral vasodilators	**nitroprusside** (NO), **fenoldopam** (dopamine receptor)	relax arterioles, given IV	**hypertensive emergencies**	–
CCBs: dihydropyridines (-dipine)	**amlodipine**, **felodipine**, **nifedipine**, **nicardipine**	block **Ca2+ influx** into arterial smooth muscle	**first line**	–
CCBs: non-dihydropyridines	**verapamil**, **diltiazem**	block Ca2+ influx; also **slow the SA node**	**first line**	–
ACE inhibitors (-pril)	**captopril**, **enalapril**, **ramipril**	block **ACE** → ↓ Ang II, ↑ **bradykinin**	**first line**; **CKD** (↓ proteinuria), esp. **diabetics**	**dry cough**, **angioedema**, **not in pregnancy**
ARBs (-sartan)	**losartan**, **valsartan**, **candesartan**, **irbesartan**, **telmisartan**	block the **AT1** receptor; **no bradykinin** effect	**first line**; when an ACEi causes cough	not in pregnancy
Renin inhibitor (-kiren)	**aliskiren**	blocks **renin** (top of the RAAS)	hypertension	–

SUMMARY TABLE: BLOOD PRESSURE CATEGORIES			
Category	**Systolic**		**Diastolic**
Normal	below 120	**and**	below 80
Elevated	120–129	**and**	below 80
Hypertension stage 1	130–139	**or**	80–89
Hypertension stage 2	140 or higher	**or**	90 or higher
Hypertensive crisis	above 180	**and/or**	above 120

END OF LESSON